1: Login and Verification.

```
abdi — root@assignment: ~ — ssh root@143.110.224.135 -p 22 — 85×31
The authenticity of host '143.110.224.135 (143.110.224.135)' can't be established.
ED25519 key fingerprint is SHA256:opB65E94v+EMKS6KpWg1bThFrnRlUvJ+vIBXKZbGvbc.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '143.110.224.135' (ED25519) to the list of known hosts.
root@143.110.224.135's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-71-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Mon Mar 23 05:02:05 UTC 2026

  System load:  0.32               Processes:             101
  Usage of /:   4.4% of 47.39GB    Users logged in:       0
  Memory usage: 10%                IPv4 address for eth0: 143.110.224.135
  Swap usage:   0%                 IPv4 address for eth0: 10.48.0.5

Expanded Security Maintenance for Applications is not enabled.

166 updates can be applied immediately.
109 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


Last login: Sun Mar 22 21:09:14 2026 from 154.153.231.210
root@assignment:~#
```

```
abdi — root@assignment: ~/Aomar_data_engineering — ssh root@143.110.224.135 -p 22 — 85×25
root@assignment:~# pwd
/root
root@assignment:~# cd Aomar_data_engineering/
root@assignment:~/Aomar_data_engineering# pwd
/root/Aomar_data_engineering
root@assignment:~/Aomar_data_engineering# whoami
root
root@assignment:~/Aomar_data_engineering#
```

Part 2: Folder and File Creation.

```
root@assignment:~# mkdir Aomar_data_engineering
root@assignment:~# cd Aomar_data_engineering/
root@assignment:~/Aomar_data_engineering# mkdir raw_data processed_data logs scripts
root@assignment:~/Aomar_data_engineering# cd raw_data/
root@assignment:~/Aomar_data_engineering/raw_data# touch products_raw_data.csv
root@assignment:~/Aomar_data_engineering/raw_data# cd ..
root@assignment:~/Aomar_data_engineering# cd processed_data/
root@assignment:~/Aomar_data_engineering/processed_data# touch products_clean.csv
root@assignment:~/Aomar_data_engineering/processed_data# cd ..
root@assignment:~/Aomar_data_engineering# cd logs/
root@assignment:~/Aomar_data_engineering/logs# touch index.log
root@assignment:~/Aomar_data_engineering/logs# cd ..
root@assignment:~/Aomar_data_engineering# touch scripts/
root@assignment:~/Aomar_data_engineering# ls
logs  processed_data  raw_data  scripts
root@assignment:~/Aomar_data_engineering# cd scripts/
root@assignment:~/Aomar_data_engineering/scripts# ls
root@assignment:~/Aomar_data_engineering/scripts# touch scripts.py
root@assignment:~/Aomar_data_engineering/scripts# ls
scripts.py
root@assignment:~/Aomar_data_engineering/scripts#
```

Part 3: File Operations.

```
root@assignment:~/Aomar_data_engineering/scripts# cd ..
root@assignment:~/Aomar_data_engineering# ls
logs  processed_data  raw_data  scripts
root@assignment:~/Aomar_data_engineering# cd raw_data
root@assignment:~/Aomar_data_engineering/raw_data# ls
products_raw_data.csv
root@assignment:~/Aomar_data_engineering/raw_data# cp products_raw_data.csv products_
raw_data_backup.csv
root@assignment:~/Aomar_data_engineering/raw_data# ls
products_raw_data.csv  products_raw_data_backup.csv
root@assignment:~/Aomar_data_engineering/raw_data# cd ..
root@assignment:~/Aomar_data_engineering# cp raw_data/products_raw_data.csv processed
_data/
root@assignment:~/Aomar_data_engineering# cd processed_data/
root@assignment:~/Aomar_data_engineering/processed_data# ls
products_clean.csv  products_raw_data.csv
root@assignment:~/Aomar_data_engineering/processed_data# mv products_raw_data.csv pro
ducts_raw_data_v2.csv
root@assignment:~/Aomar_data_engineering/processed_data# cd ../raw_data/
root@assignment:~/Aomar_data_engineering/raw_data# ls
products_raw_data.csv  products_raw_data_backup.csv
root@assignment:~/Aomar_data_engineering/raw_data# rm products_raw_data.csv
root@assignment:~/Aomar_data_engineering/raw_data# ls
products_raw_data_backup.csv
root@assignment:~/Aomar_data_engineering/raw_data#
```

Part 4: Permissions

```
435  cp raw_data/products_raw_data.csv processed_data/
436  cd processed_data/
437  ls
438  mv products_raw_data.csv products_raw_data_v2.csv
439  cd ../raw_data/
440  ls
441  rm products_raw_data.csv
442  ls
443  cd ..
444  ls
445  ls -l
446  chmod -r 700 Aomar_data_engineering/
447  chmod -R 700 Aomar_data_engineering/
448  ls -l
```

```
abdi — root@assignment: ~/Aomar_data_engineering/scripts — ssh root@143.110.224.135 -p 22 — 85×27
total 16
drwx------ 2 root root 4096 Mar 23 05:07 logs
drwx------ 2 root root 4096 Mar 23 05:14 processed_data
drwx------ 2 root root 4096 Mar 23 05:15 raw_data
drwx------ 2 root root 4096 Mar 23 05:08 scripts
root@assignment:~/Aomar_data_engineering# ls -la
total 24
drwx------  6 root root 4096 Mar 23 05:05 .
drwx------ 22 root root 4096 Mar 24 00:22 ..
drwx------  2 root root 4096 Mar 23 05:07 logs
drwx------  2 root root 4096 Mar 23 05:14 processed_data
drwx------  2 root root 4096 Mar 23 05:15 raw_data
drwx------  2 root root 4096 Mar 23 05:08 scripts
root@assignment:~/Aomar_data_engineering# cd processed_data/
root@assignment:~/Aomar_data_engineering/processed_data# ls -la
total 8
drwx------ 2 root root 4096 Mar 23 05:14 .
drwx------ 6 root root 4096 Mar 23 05:05 ..
-rwx------ 1 root root    0 Mar 23 05:06 products_clean.csv
-rwx------ 1 root root    0 Mar 23 05:13 products_raw_data_v2.csv
root@assignment:~/Aomar_data_engineering/processed_data# cd ../scripts/
root@assignment:~/Aomar_data_engineering/scripts# ls -la
total 8
drwx------ 2 root root 4096 Mar 23 05:08 .
drwx------ 6 root root 4096 Mar 23 05:05 ..
-rwx------ 1 root root    0 Mar 23 05:08 scripts.py
root@assignment:~/Aomar_data_engineering/scripts#
```

Part 5: Navigation and History.

```
abdi — root@assignment: ~/Aomar_data_engineering/scripts — ssh root@143.110.224.135 -p 22 — 85×27
total 16
drwx------ 2 root root 4096 Mar 23 05:07 logs
drwx------ 2 root root 4096 Mar 23 05:14 processed_data
drwx------ 2 root root 4096 Mar 23 05:15 raw_data
drwx------ 2 root root 4096 Mar 23 05:08 scripts
root@assignment:~/Aomar_data_engineering# ls -la
total 24
drwx------  6 root root 4096 Mar 23 05:05 .
drwx------ 22 root root 4096 Mar 24 00:22 ..
drwx------  2 root root 4096 Mar 23 05:07 logs
drwx------  2 root root 4096 Mar 23 05:14 processed_data
drwx------  2 root root 4096 Mar 23 05:15 raw_data
drwx------  2 root root 4096 Mar 23 05:08 scripts
root@assignment:~/Aomar_data_engineering# cd processed_data/
root@assignment:~/Aomar_data_engineering/processed_data# ls -la
total 8
drwx------ 2 root root 4096 Mar 23 05:14 .
drwx------ 6 root root 4096 Mar 23 05:05 ..
-rwx------ 1 root root    0 Mar 23 05:06 products_clean.csv
-rwx------ 1 root root    0 Mar 23 05:13 products_raw_data_v2.csv
root@assignment:~/Aomar_data_engineering/processed_data# cd ../scripts/
root@assignment:~/Aomar_data_engineering/scripts# ls -la
total 8
drwx------ 2 root root 4096 Mar 23 05:08 .
drwx------ 6 root root 4096 Mar 23 05:05 ..
-rwx------ 1 root root    0 Mar 23 05:08 scripts.py
root@assignment:~/Aomar_data_engineering/scripts#
```

```
 1237  history
root@assignment:~# !!
history
  238  cd ..
  239  rm raw_data/sales_raw_backup.csv
  240  cd raw_data
  241  ls
  242  cd ..
  243  ls -R
```

Article

https://dev.to/abdiomari/linux-fundamentals-for-data-engineering-4gh2